5. Subnetting and CIDR (with practice)

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Phase: Phase 2 — Addressing and subnets
Estimated time: 2–3 days
Slug: `05-subnetting`

Learning objectives

- Convert prefix lengths to masks
- Compute network/broadcast/usable hosts
- Choose subnets for labs
- Use `ipcalc`/`sipcalc` or manual method

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

CIDR quick facts

`192.168.10.0/24` → mask `255.255.255.0` → 256 addresses → 254 usable (typical).

Host bits = 32 - prefix. Usable ≈ 2^hostbits − 2 (except some special cases).

```
# If installed
ipcalc 192.168.10.0/24
sipcalc 10.0.0.0/12

# Python one-liner helper (lab)
python3 - <<'PY'
import ipaddress
n=ipaddress.ip_network('192.168.10.0/24')
print(n.network_address, n.broadcast_address, n.num_addresses)
PY
```

Common prefixes

Prefix	Hosts (approx usable)	Typical use
/32	1	single host route
/30	2	point-to-point
/24	254	small LAN
/16	~65k	large private
/0	default	everything

Common mistakes

- Off-by-one in usable host counts
- Overlapping subnets in VPCs
- Using /24 everywhere without design

Exercises

Work in `exercises/05-subnetting/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/05-subnetting/` until after an honest attempt.

Self-check

1. How many host bits in /26?
2. What does /32 mean?
3. Why avoid overlapping CIDRs?

Further reading

- https://www.rfc-editor.org/rfc/rfc4632